

# Substation ESG Report

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks. Updated annually via the SSI Index scoring engine.

SUBSTATION

**Substation Example**

MX\_001\_000 · Ciudad de México, Ciudad de México

SSI SCORE ( $R_{\text{MEDIAN}}$ )

**0.323**

● Low Risk

50.0th percentile

VOLTAGE CLASS

**230 kV**

High-voltage transmission

CONFIDENCE INTERVAL

**0.207 – 0.429**

P5–P95 · CI width 0.223 · medium confidence

MARKOV ETTC

**18.5 yr**

Expected time to criticality

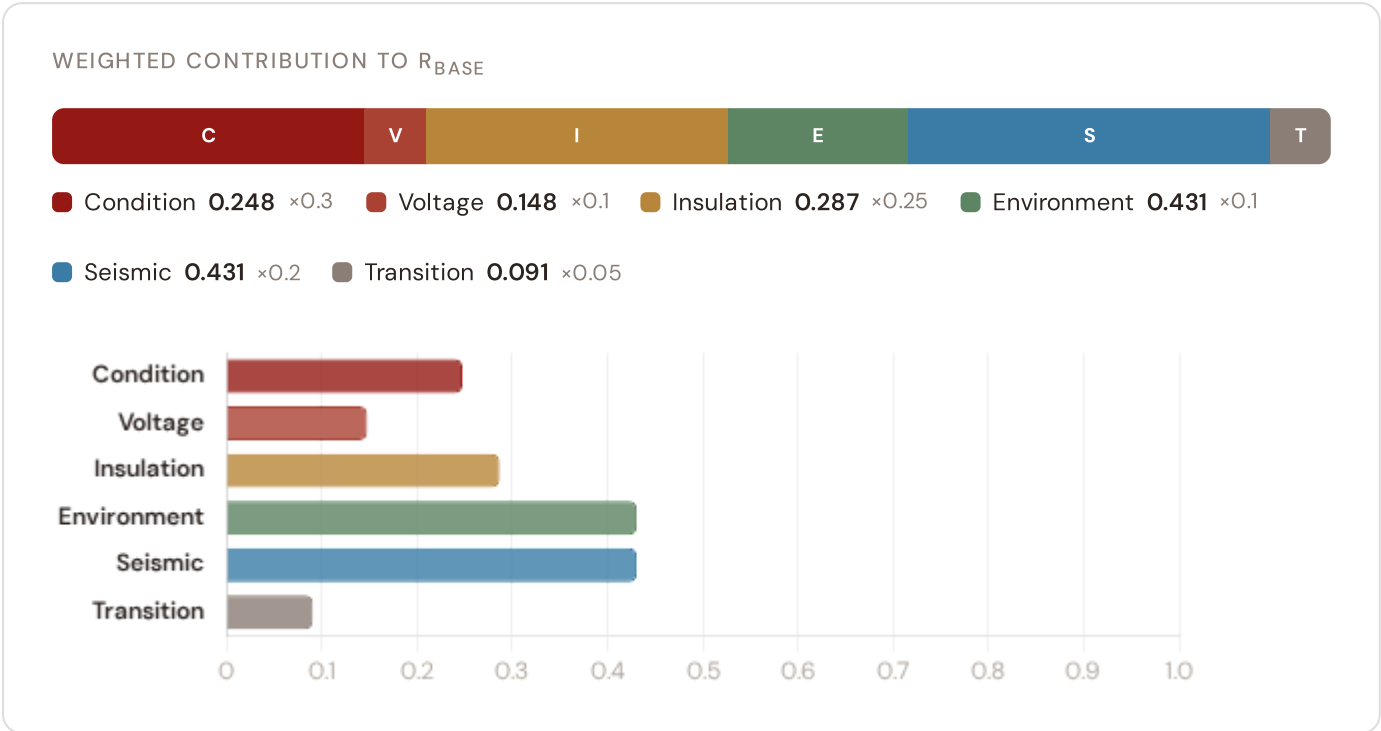
DER PENETRATION

**0.12**

T1 transition stress score

## Component Fingerprint

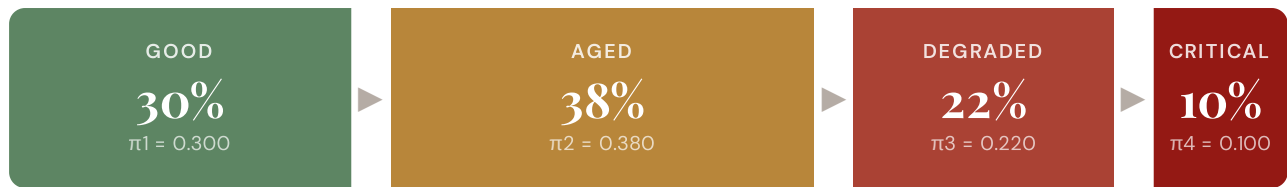
Six-component weighted decomposition of the SSI composite score — the substation's structural risk profile



## Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

#### STEADY-STATE DISTRIBUTION



Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

#### RISK SCORE

**0.386**

Composite Markov risk

#### ETTC

**18.5 yr**

Expected time to critical

#### P(CRITICAL) 20YR

**10.0%**

Probability of reaching critical

#### CORROSION

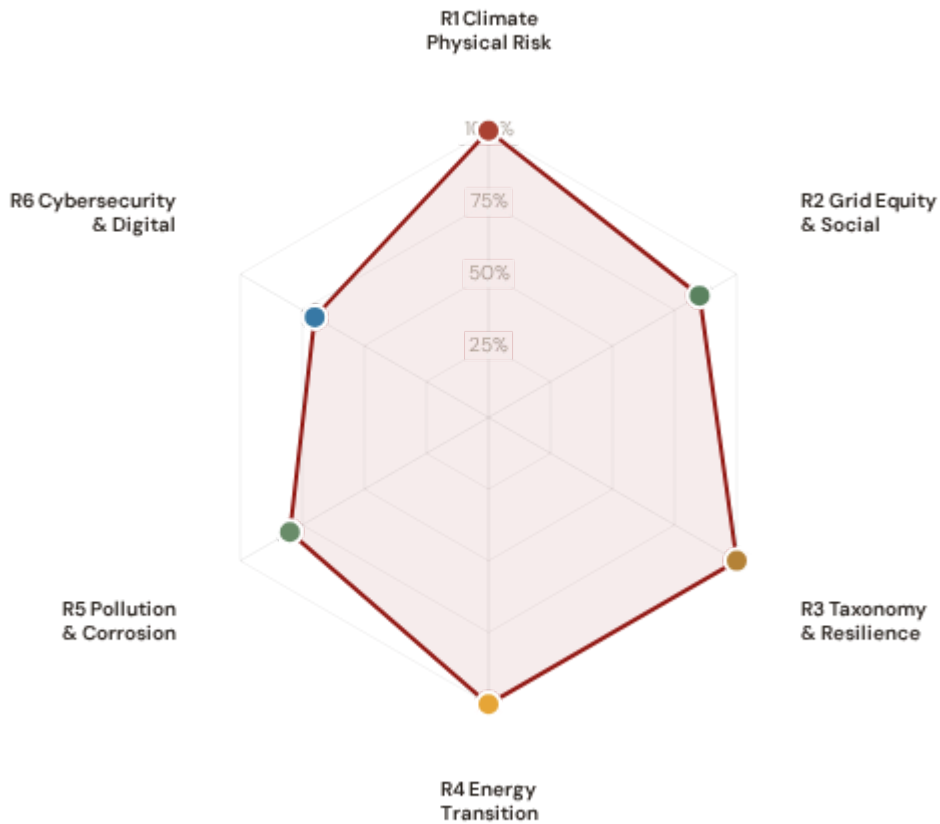
**C1**

ISO 9223 classification



## ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals



● **R1 · SSI Annual Risk Report — Mexico Fleet** READY

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13

SDG 9

SDG 11

● **R2 · Regional Deep-Dive: Gulf Coast Corridor** READY

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7

SDG 10

SDG 1

● **R3 · Hurricane & Seismic Resilience Assessment** READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9

SDG 13

SDG 12

● **R4 · Energy Transition Scorecard** READY

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7

SDG 13

SDG 9

● **R5 · Infrastructure Modernization Tracker** READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11

SDG 3

SDG 15

## SSI Annual Risk Report — Mexico Fleet

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

### WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure across Mexico's 32 estados at substation level using ERA5 reanalysis and Markov degradation modelling.

### SDG Alignment

#### SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. Mexico's dual hazard exposure (hurricanes on Gulf+Pacific coasts, seismic activity along Cocos Plate, volcanic activity) makes this assessment critical. With a Markov risk score of 0.386 and ETTC of 18.5 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13

SDG 9

SDG 11

## Substation Profile

|                               |            |
|-------------------------------|------------|
| Insulation Risk (I component) | 0.287      |
| Environment Component (E)     | 0.431      |
| Markov Risk Score             | 0.386      |
| ETTC (time to criticality)    | 18.5 years |
| P(critical) at 20 years       | 10.0%      |
| Corrosion Class               | C1         |
| Seismic PGA (g)               | 0.1750     |
| Climate $\Delta$ R 2050       | 0.1130     |

## SSI Variables Feeding This Report

| SSI VARIABLE                      | VALUE  | ESG DISCLOSURE REQUIREMENT                       | STATUS |
|-----------------------------------|--------|--------------------------------------------------|--------|
| I1 Storm/Hurricane IRI            | 0.115  | ESRS E1-9: Financial effects from physical risks | READY  |
| I2 Seismic/Volcanic IRI           | 0.086  | TCFD Strategy (b): Physical risk impact          | READY  |
| I3 Heat-wave IRI                  | 0.057  | EU Taxonomy Annex A: Heat stress                 | READY  |
| I5 Thermal stress                 | 0.029  | ESRS E1-4: Climate mitigation targets            | READY  |
| Markov Risk score                 | 0.386  | TCFD Risk Management: Risk identification        | READY  |
| P_crit P(critical) 20yr           | 10.0%  | ESRS E1-9: Time horizons                         | READY  |
| R6b Seismic PGA                   | 0.1750 | TCFD Strategy (b): Seismic physical risk         | READY  |
| $\Delta$ R2050 Climate trajectory | 0.1130 | ESRS E1-9: Forward-looking climate impact        | READY  |

Data Sources & Vintage

|                                                                                               |                |
|-----------------------------------------------------------------------------------------------|----------------|
| ERA5 Climate Reanalysis<br>Copernicus CDS                                                     | 2024 Weekly    |
| Hurricane Data (East Pacific, Atlantic)<br>SMN (Servicio Meteorológico Nacional)              | 2024 Seasonal  |
| Seismic Data (Cocos Plate)<br>SGM (Servicio Geológico Mexicano)                               | 2024 Real-time |
| Volcanic Risk (Popocatepetl, Colima)<br>CENAPRED (Centro Nacional de Prevención de Desastres) | 2023 Annual    |
| Weather & Climate Data<br>SMN (Servicio Meteorológico Nacional)                               | 2024 Daily     |
| IEEE C57.91 Thermal Model<br>IEEE                                                             | Standard N/A   |
| CIGRE TB 761 Markov<br>CIGRE                                                                  | 2019 N/A       |
| CMIP6 SSP2-4.5 Projections<br>Copernicus CDS                                                  | — Not ingested |

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. SMN weather data informs hurricane and seismic vulnerability assessment. Markov 5–state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. All inputs are production–grade with institutional provenance from CENACE, CRE, INEGI, and SGM.

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Regional Deep-Dive: Gulf Coast Corridor

READY

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. This report focuses on Mexico's vulnerability corridor: Veracruz, Tabasco, and Tamaulipas. These states face the highest hurricane exposure and lowest per-capita GDP, creating a dual exposure: physical infrastructure risk + energy poverty. The SSI Index is uniquely positioned to surface where grid infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V\_socio of 0.15. The R3 social modifier (1.049) amplifies risk scores in regions with high unemployment, poverty concentration, and energy poverty — making spatial inequality visible and quantifiable at asset level. In Mexico's hurricane-vulnerable states, this correlation is particularly acute.

SDG 7

SDG 10

SDG 1

Substation Profile

|                                |             |
|--------------------------------|-------------|
| V_socio (energy poverty index) | 0.15        |
| EP Rate (energy poverty %)     | 0.12%       |
| Unemployment Rate              | 3.5%        |
| GDP per Capita                 | MXN 132,000 |
| R&D % of GDP                   | 0.015%      |
| R3 Social Multiplier           | 1.049       |
| Poverty Rate                   | 16.8%       |
| Indigenous Population          | 2%          |
| Elderly Population %           | 0.12%       |
| Catchment Population           | 45,840      |

### SSI Variables Feeding This Report

| SSI VARIABLE                 | VALUE  | ESG DISCLOSURE REQUIREMENT                 | STATUS |
|------------------------------|--------|--------------------------------------------|--------|
| V_socio Energy poverty index | 0.15   | ESRS S2-4: Impacts on affected communities | READY  |
| EP_rate Energy poverty rate  | 0.12%  | SFDR PAI #14: Fossil fuel exposure proxy   | READY  |
| R3 Social multiplier         | 1.049  | ESRS S1-6: Workforce characteristics       | READY  |
| Elderly Elderly population % | 0.12%  | ESRS S2: Vulnerable populations            | READY  |
| Pop Catchment population     | 45,840 | ESRS S1: Affected communities              | READY  |

### Data Sources & Vintage

|                                                                                 |             |
|---------------------------------------------------------------------------------|-------------|
| Population & Economics<br>INEGI (Instituto Nacional de Estadística y Geografía) | 2023 Annual |
| Energy Market Data<br>CENACE (Centro Nacional de Control de Energía)            | 2024 Annual |
| Grid Reliability & DSO Data<br>CRE (Comisión Reguladora de Energía)             | 2023 Annual |

#### TECHNICAL VALIDITY

V\_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using INEGI household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, poverty concentration, and energy poverty. Regional data sourced from INEGI, CONEVAL, and CFE distribution company reports.

3

## Hurricane & Seismic Resilience Assessment

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

READY

### WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. Mexico faces dual extreme hazards: (1) Atlantic and Pacific hurricanes creating wind and storm surge risk on Gulf Coast (Veracruz, Tabasco) and Pacific coast (Jalisco, Nayarit), and (2) seismic activity along the Cocos Plate subduction zone affecting central states. The SSI composite score (R\_median), 6 components, modifier



architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R\_median (0.323), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting. Mexico's infrastructure modernization imperative (Modernizador de la Red Eléctrica, MRE) depends on such systematic resilience assessment.

SDG 9

SDG 13

SDG 12

Substation Profile

|                        |                    |
|------------------------|--------------------|
| R_median (composite)   | 0.323              |
| R_base (pre-modifier)  | 0.329              |
| Modifier Impact        | -1.69%             |
| C (Condition)          | 0.248              |
| V (Voltage)            | 0.148              |
| I (Insulation/Climate) | 0.287              |
| E (Environment)        | 0.431              |
| S (Seismic)            | 0.431              |
| T (Transition)         | 0.091              |
| Seismic PGA (g)        | 0.1750 (Moderate)  |
| Climate Δ R 2050       | +0.1130 (SSP2-4.5) |

SSI Variables Feeding This Report

| SSI VARIABLE             | VALUE       | ESG DISCLOSURE REQUIREMENT                    | STATUS |
|--------------------------|-------------|-----------------------------------------------|--------|
| R_median Composite score | 0.323       | EU Taxonomy Art. 11: Adaptation screening     | READY  |
| 6 Comp. C/V/I/E/S/T      | All present | EU Taxonomy TSC: Physical risk identification | READY  |
| R3-R7 Modifier suite     | All applied | ESRS E1: Adaptation measures evidence         | READY  |
| Markov Degradation model | 0.386       | TCFD: Forward-looking risk assessment         | READY  |
| R5 Asymmetric CI         | 0.223       | ESRS: Uncertainty quantification              | READY  |
| PGA Seismic hazard       | 0.1750      | EU Taxonomy: Seismic risk screening           | READY  |

Data Sources & Vintage

|                                                                                  |               |
|----------------------------------------------------------------------------------|---------------|
| ERA5 Climate Reanalysis<br>Copernicus CDS                                        | 2024 Weekly   |
| Hurricane Data (East Pacific, Atlantic)<br>SMN (Servicio Meteorológico Nacional) | 2024 Seasonal |
| Population & Economics<br>INEGI (Instituto Nacional de Estadística y Geografía)  | 2023 Annual   |
| CIGRE TB 761 Markov<br>CIGRE                                                     | 2019 N/A      |

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. Hazard inputs sourced from SMN (hurricane/weather), SGM (seismic), and CENAPRED (multi-hazard risk assessment).

WHY THIS REPORT EXISTS

Mexico's energy transition is bifurcated: solar + wind expansion (especially in the north and Istmo de Tehuantepec) creates distributed generation stress on the grid, while fossil fuel retirement (coal, fuel oil) reallocates transmission flows. The SIN (Sistema Interconectado Nacional, ~95% of Mexico's grid) must manage this while maintaining service to 32 estados. The T component measures grid stress from renewable integration and EV charging proliferation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. Mexico's renewable energy target: 35% by 2024, 50% by 2030. This substation has a DER penetration of 12.2%. The T1\_score (0.091) measures the stress this places on the substation.

SDG 7

SDG 13

SDG 9

Substation Profile

|                              |          |
|------------------------------|----------|
| T1 Score (transition stress) | 0.091    |
| DER Penetration              | 12.2%    |
| Solar + Wind Capacity        | 196.3 MW |
| EV Fleet Share               | 1.2%     |
| RE Share (local)             | 15.8%    |
| T Component Weight           | 0.05     |

SSI Variables Feeding This Report

| SSI VARIABLE               | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|----------------------------|-------|-------------------------------------|--------|
| T1 Transition stress score | 0.091 | ESRS E1: Transition plan assessment | READY  |
| DER DER penetration        | 12.2% | TCFD Transition: Technology risk    | READY  |
| Solar+Wind Capacity (MW)   | 196.3 | Mexico Green Deal: Grid flexibility | READY  |
| EV EV fleet share          | 1.2%  | SDG 7: Clean energy access          | READY  |

Data Sources & Vintage

|                                                                      |              |
|----------------------------------------------------------------------|--------------|
| Energy Market Data<br>CENACE (Centro Nacional de Control de Energía) | 2024 Annual  |
| Grid Reliability & DSO Data<br>CRE (Comisión Reguladora de Energía)  | 2023 Annual  |
| Renewable Installations<br>CENACE SIN Database                       | 2024 Monthly |
| Distribution Data<br>CFE (Comisión Federal de Electricidad)          | 2024 Monthly |

TECHNICAL VALIDITY

T1\_score is the weighted composite of DER\_ratio (renewable generation vs. local demand), DER\_variability (intermittency), and EV\_load\_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from CENACE (Sistema de Información Operacional del SIN) and CFE distribution company reports.

5

Infrastructure Modernization Tracker

READY

ESRS E2 Pollution · ISO 9223 Corrosion Classification · Environmental Impact

WHY THIS REPORT EXISTS

Mexico's grid infrastructure is aging: many substations date to the 1970s–1990s. Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. Mexico's coastal substations (Veracruz, Tamaulipas) face elevated salt-spray corrosion; highland substations (Popocatepetl, Colima volcanic zones) face atmospheric chemistry stress. The ISO 9223 classification captures this exposure.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C1) and E component (0.431) track environmental degradation and pollution sensitivity at asset level. CENACE and CFE modernization programmes (MRE) depend on such systematic assessment to prioritize upgrade pathways.

SDG 11

SDG 3

SDG 15

### Substation Profile

|                            |       |
|----------------------------|-------|
| Corrosion Class (ISO 9223) | C1    |
| E2 Local Enrichment        | 0.43  |
| E Component Score          | 0.431 |
| Markov Risk Score          | 0.386 |

### SSI Variables Feeding This Report

| SSI VARIABLE                       | VALUE | ESG DISCLOSURE REQUIREMENT             | STATUS |
|------------------------------------|-------|----------------------------------------|--------|
| Corrosion ISO 9223 class           | C1    | ESRS E2: Pollution risk classification | READY  |
| E2_local Environmental sensitivity | 0.43  | ESRS E2: Local environmental impact    | READY  |
| E Environment component            | 0.431 | ISO 9223: Atmospheric corrosion        | READY  |

### Data Sources & Vintage

|                                                                                          |              |
|------------------------------------------------------------------------------------------|--------------|
| Environmental Data<br>SEMARNAT (Secretaría de Medio Ambiente)                            | 2023 Annual  |
| Air Quality Monitoring<br>CONUEE (Comisión Nacional para el Uso Eficiente de la Energía) | 2023 Monthly |
| ISO 9223 Corrosion<br>ISO                                                                | 2012 N/A     |

#### TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Mexico's coastal climate and volcanic gas exposure in central states produce elevated corrosion in coastal and high–altitude zones. Full validation requires SEMARNAT air quality monitoring overlay at substation coordinates, plus SMN weather station data.

WHY THIS REPORT EXISTS

Mexico's 3,140 substations serve a population of 128 million across 32 estados and 2,457 municipios. Energy poverty varies dramatically: urban centers enjoy near-universal access, rural areas lag. Cybersecurity is an emerging challenge: grid digitalization under CENACE/CRE modernization initiatives exposes legacy SCADA systems. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national cyber maturity to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (1.017) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. For Mexico, with 3,140 substations and 30,396 power lines serving 128 million people across 32 estados, systemic cyber vulnerability poses an economy-wide energy security risk.

SDG 9

SDG 16

Substation Profile

|                            |        |
|----------------------------|--------|
| R7 Cyber Modifier          | 1.017  |
| Graph Degree (topology)    | 6      |
| BC Percentile (centrality) | 0.400  |
| Is Bridge Node             | No     |
| Cluster Coefficient        | 0.0130 |

SSI Variables Feeding This Report

| SSI VARIABLE              | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|---------------------------|-------|-------------------------------------|--------|
| R7 Cyber modifier         | 1.017 | NIS2: Cybersecurity risk management | READY  |
| BC Betweenness centrality | 0.400 | ESRS G1: Governance & topology risk | READY  |
| Degree Graph degree       | 6     | Grid resilience: Connectivity       | READY  |

Data Sources & Vintage

Distribution Data

CFE (Comisión Federal de Electricidad)

2024 Monthly

Cybersecurity Index

ENISA (adapted for Mexico)

2024 Biennial

TECHNICAL VALIDITY

R7\_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators adapted for Mexico, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. CRE and CENACE cyber audit data inform this component.

C

## Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

| ITEM                 | VALUE                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------|
| Scoring Engine       | SSI Index v4.0.2 — Gaussian copula Monte Carlo (10,000 iterations)                            |
| Weight Architecture  | C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05                                                |
| Modifiers Applied    | R3 C_mult: 1.049, R4 F_topo: 1.123, R6 restoration: 1.030, R6 seismic: 1.051, R7 cyber: 1.017 |
| Degradation Model    | Markov 5-state, CIGRE TB 761 calibration                                                      |
| Thermal Model        | IEEE C57.91 transformer loading curves                                                        |
| Report Date          | 17 March 2026                                                                                 |
| Reporting Period     | Calendar year 2025                                                                            |
| Refresh Cadence      | Annual (data ingestion → scoring engine → display)                                            |
| Substation           | Substation Example (MX_001_000)                                                               |
| Data Assurance Level | All inputs from institutional open-source with citable vintage                                |

